

5.3.7 Wake-up time from low-power modes and voltage scaling transition times

The wake-up times given in the table below are the latency between the event and the execution of the first user instruction.

The device goes in low-power mode after the WFE (wait for event) instruction

Table 29. Wake-up time from low-power modes

1. Evaluated by characterization - Not tested in production.

2. The wakeup times are measured from the wakeup event to the point in which the application code reads the first instruction.

Symbol	Parameter	Conditions	Wakeup clock	f _{HCLK} (MHz)	Typ	Unit
t _{wu} (Sleep)	Wakeup from Sleep	Instruction cache enabled or disabled, HSI 48 MHz	-	-	16	CPU clock cycles
		Instruction cache enabled or disabled, HSI 144 MHz				
t _{wu} (Stop)	Wakeup from Stop 0	Flash memory in normal mode	HSI	144	3.85	μs
		Flash memory in low-power mode	HSI	144	7.55	
		Flash memory in normal mode	HSI DIV3	48	5.65	
		Flash memory in low-power mode	HSI DIV3	48	9.20	
	Wakeup from Stop 1	Flash memory in normal mode	HSI	144	37.50	
		Flash memory in low-power mode	HSI	144	41.00	
		Flash memory in normal mode	HSI DIV3	48	39.00	
		Flash memory in low-power mode	HSI DIV3	48	43.00	
t _{wu} (Standby)	Wakeup from Standby mode	-	-	-	445	

Table 30. Wake-up time using USART/LPUART

Symbol	Parameter	Condition	Typ	Max ⁽¹⁾	Unit
t _{wu} USART/ t _{wu} LPUART	Wake-up time needed to calculate the maximum USART/LPUART baudrate allowing to wake up from Stop mode when USART/LPUART clock source is 48 MHz by HSI DIV3	Stop 0 mode	4.8	6.6	μs
		Stop 1 mode			

1. Specified by design - Not tested in production.

5.3.8 External clock timing characteristics

5.3.8.1 High-speed external user clock generated from an external source

In bypass mode, the HSE oscillator is switched off and the input pin is a standard GPIO.

The external clock signal has to respect the I/O characteristics in I/O port characteristics. However, the recommended clock input waveform is shown in the figure below.

Table 31. High-speed external user clock characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f _{HSE_ext}	User external clock source frequency	Digital mode (HSEYBYP = 1, HSEEXT = 1)	-	-	50	MHz
		Analog mode (HSEYBYP = 1, HSEEXT = 0)	4	-	50	
V _{HSEH}	OSC_IN input pin high-level voltage	Digital mode (HSEYBYP = 1, HSEEXT = 1)	0.7 × V _{DD}	-	V _{DD}	V
V _{HSEL}	OSC_IN input pin low-level voltage	Digital mode (HSEYBYP = 1, HSEEXT = 1)	V _{SS}	-	0.3 × V _{DD}	
t _w (HSEH) t _w (HSEL)	OSC_IN high or low time	Digital mode (HSEYBYP = 1, HSEEXT = 1)	7	-	-	ns